# Classical integrability

## § Motivation / Poisson brackets.

- Newton: $m\ddot{\vec{y}}_i = \vec{F}_i(\{\vec{y}_j\}, \{\dot{\vec{y}}_j\})$. Often $\longrightarrow -\text{grad}(V(\vec{y}_i))$.
- Observation: Often $\vec{F}_i$'s are recovered from $H(\{\vec{y}_j\}, \{\dot{\vec{y}}_j\})$ - Energy., Energy is conserved.
- For us: we want a map

$$V_\bullet : \mathcal{F}(M) \longrightarrow \text{Vect}(M) = \text{Der}(\mathcal{F}(M))$$
$$f \longmapsto V_f$$

Such that $V_f(f) = 0$.

Here $M \longleftrightarrow \{\{(\vec{y}_j, \dot{\vec{y}}_j = \vec{p}_j)\}\}$ - configuration space

$\text{Vect}(M) \longleftrightarrow$ "Forces" = Dynamics on $M$ = ODE on $M$ (first order)

- Equivalently, could write

$$P : \mathcal{F}(M) \otimes \mathcal{F}(M) \longrightarrow \mathcal{F}(M)$$
$$f \otimes g \longmapsto V_f(g)$$

Such that $V_f(g_1 g_2) = V_f(g_1) g_2 + g_1 V_f(g_2)$

And $V_f(f) = 0 \ \forall f \Rightarrow$ ~~$P((f+g)\otimes(f+g)) = 0$~~

$$\cancel{P(f\otimes f)} + P(f\otimes g) + P(g\otimes f) + \cancel{P(g\otimes g)}$$

$\Rightarrow P$ is skew symmetric.

Denote further $P(f\otimes g) = \{f, g\}$.

Let us also ask for

$$V_{\{f,g\}} = [V_f, V_g] \Leftrightarrow \{\{f,g\},h\} = \{f,\{g,h\}\} - \{g,\{f,h\}\} \Leftrightarrow$$

$\Leftrightarrow$ Jacobi identity on $\{\cdot,\cdot\}$.

**Def** bi-derivation $\{\cdot,\cdot\}: \mathcal{F}(M)\otimes\mathcal{F}(M)\longrightarrow\mathcal{F}(M)$ & Lie algebra

Is called a Poisson bracket.

- **Remark**: Sufficient to define $\{\cdot,\cdot\}$ on coordinate ("linear") functions then bi-derivation continues uniquely. Need to check Lie algebra. Namely, if $(x_1,\dots x_n)$-local coords $\Rightarrow \{f,g\}=\sum_{i,j}\partial_i(f)\partial_j(g)\{x_i,x_j\}$.

**Examples**

1) Let $M=\mathbb{R}^{2n}=\{(x_1,\dots x_n,p_1,\dots p_n)\}$, $\mathcal{F}(M)=C^\infty(\mathbb{R}^{2n})$.

Set $\{f,g\}(x_1,\dots x_n,p_1,\dots p_n)=-\sum_{i=1}^{n}\left(\frac{\partial f}{\partial x_i}\frac{\partial g}{\partial p_i}-\frac{\partial f}{\partial p_i}\frac{\partial g}{\partial x_i}\right)\Leftrightarrow$ $\{x_i,x_j\}=0$, $\{p_i,p_j\}=0$, $\{x_i,p_j\}=\delta_{ij}$

2) Let $M=\mathbb{R}^m=\{(x_1,\dots x_m)\}$, $\mathcal{F}(M)=C^\infty(\mathbb{R}^m)$

Fix $\epsilon_{ij}\in Mat_{m\times m}(\mathbb{R})$, skew-symm.

Set $\{f,g\}(x_1,\dots x_m)=\sum_{i,j=1}^{m}\epsilon_{ij}x_ix_j\frac{\partial f}{\partial x_i}\frac{\partial g}{\partial x_j}$.

3) $\mathfrak{g}$ is a Lie algebra (f.d.) [KKS bracket].

$M=\mathfrak{g}^*=\{(e_1,\dots e_n)\}$, $\mathcal{F}(M)=\mathbb{C}[e_1,\dots e_n]=S^\bullet(\mathfrak{g})$.

$(e_i\in\mathfrak{g})$

Set $\forall v,w\in\mathfrak{g}\subset S^\bullet(\mathfrak{g})=\mathcal{F}(M)$

$\{v,w\}=[v,w]$.

Or, $\{f,g\}=\sum_{i,j=1}^{n}\frac{\partial f}{\partial e_i}\frac{\partial g}{\partial e_j}[e_i,e_j]=\sum_{i,j,k=1}^{n}\frac{\partial f}{\partial e_i}\frac{\partial g}{\partial e_j}c_{ij}^{k}e_k$ ($c_{ij}^k$: structure constants.)

- **Remark** $\{f,g\}(x)$ depends only on $df,dg$ at $x$. Write $\{f,g\}(x)=\Pi_x(df,dg)$ $\Pi_x\in\Lambda^2T_xM$.

The Poisson bracket is called non-degenerate at $x\in M$ if $\Pi_x: T_x^*M\otimes T_x^*M\longrightarrow\mathbb{R}$ is so.

~~$\forall f: df\neq 0$~~ $\forall\alpha\in T_x^*M$ (over $\mathbb{C}$): $\exists\beta\in T_x^*M: \Pi_x(\alpha,\beta)\neq 0$

**Excersise** If $\dim(M)$ is odd, any Poisson bracket on $M$ is degenerate at every point.

# Motivation (II).

Let $A$ be a filtered algebra:

$$A = \bigcup_{i=0}^{\infty} A_i, \quad A_i \subseteq A_{i+1}, \quad A_i A_j \subseteq A_{i+j}$$

such that $\forall a_1, a_2 : a_1 \in A_i, a_2 \in A_j$:

$$a_1 a_2 - a_2 a_1 \in A_{i+j-1}$$

Then $\mathrm{gr}(A) = \bigoplus_{i=0}^{\infty} A_i / A_{i-1}$ is a commutative algebra

and $\{a_1, a_2\} := \pi_{i+j-1}(\tilde{a}_1 \tilde{a}_2 - \tilde{a}_2 \tilde{a}_1)$ ($\tilde{a}_2$: a representative of $a_2$ in $A_j$.)

($\pi_{i+k} : A_k \twoheadrightarrow A_k / A_{k-1} \hookrightarrow \mathrm{gr}(A)$)

is a Poisson bracket on $\mathrm{gr}(A)$.

**Example** $A = U(\mathfrak{g})$, $\mathrm{gr}(A) = S^\bullet(\mathfrak{g}) \cong \mathbb{C}[\mathfrak{g}^*]$, $\{\cdot,\cdot\}$ is KKS bracket.

**Example** $A = \{$differential operators with polynomial coefficients on $\mathbb{C}^n\}$ $= \{\sum a_{i_1,\dots,i_k}(x_1,\dots,x_n)\partial_{i_1}\cdots\partial_{i_k} + \dots\}$

$$\mathrm{gr}(A) = \mathbb{C}[T^*\mathbb{C}^n] = \mathbb{C}[x_1,\dots,x_n,p_1,\dots,p_n]$$

$$\{x_i, x_j\} = \pi_1(x_1 x_2 - x_2 x_1) = 0$$

$$\{p_i, p_j\} = \pi_1(\partial_i \partial_j - \partial_j \partial_i) = \pi_1(0) = 0$$

$$\{p_i, x_j\} = \pi_0(\partial_i \circ x_j - x_j \circ \partial_i) = \pi_0(\delta_{ij}) = \delta_{ij}.$$

## § Integrability

Let $M$ be a mfd of dim $=2n$, $\{\cdot,\cdot\}$ be a non-degenerate Poisson bracket.
We say that $H_1, H_2 \in \mathcal{F}(M)$ are in involution if
$\{H_1, H_2\} \equiv 0$.
We will call $H_1, \dots, H_m$ (functionally) independent if
$dH_1, \dots, dH_m$ are linearly independe - at generic point on $M$.

**Lemma**

Let $H_1, \dots, H_m$ be independent, in involution.
Then $m \leq n$.

**Pf.**

: Let $x \in M$ be s.t. $dH_1, \dots, dH_m$ are independent. a

$x$.

We have non-deg. bilinear form $\Pi_x$.
Then consider $W = \langle d_x H_1, \dots, d_x H_m \rangle \subset T_x^* M$.
$\dim(W^\perp) = 2n - m$, on the other hand

$\Pi_x(d_x H_i, d_x H_j) = \{H_i, H_j\}(x) = 0 \Rightarrow W \subseteq W^\perp \Rightarrow$

$\Rightarrow \dim(W) \leq \dim(W^\perp) \Rightarrow m \leq 2n - m \Rightarrow m \leq n$ ∎

**Def.** A system of functions $H_1, \dots, H_n \in \mathcal{F}(M)$:

- $\{H_i, H_j\} = 0 \ \forall i, j$ (in involution)
- $\{dH_1, \dots, dH_n\}$ are (gen.) independent:

Is called an integrable system.

**Property:** If $\gamma$ is an integral curve of $H_i$ (say, $H_1$), then
$H_j(\gamma(t)) = \text{const}$ (i.e. $\gamma \subset$ level set of $H_i$'s).

## Examples

1) $H_j = \sum_{i=1}^{n} p_i^j$.

2) $n=2$,

$H_1 = \frac{p_1^2+p_2^2}{2} + U(x_1^2+x_2^2)$

$H_2 = L = p_1 x_2 - p_2 x_1$

Dynamics of $H_1$: $\begin{cases} \dot{x}_i = \{H_1, x_i\} = p_i \\ \dot{p}_i = 2x_i U'(x_1^2+x_2^2) \end{cases}$

Introduce coordinates: $x_1 = r\cos(\varphi)$, $x_2 = r\sin(\varphi)$

$\Rightarrow \dot{r} = \frac{p_1 x_1 + p_2 x_2}{r} = p_r$

$$\dot{p}_r = \frac{1}{r}\left(2(x_1^2+x_2^2)U'(x_1^2+x_2^2) + p_1^2+p_2^2\right) = \frac{1}{r}\left(2r^2U'(r^2) + (2H_1 - U(r^2))\right)$$

$\Rightarrow \begin{cases} \dot{r} = p_r \\ \dot{p}_r = 2rU'(r^2) + \frac{2H_1 - U(r^2)}{r} \end{cases}$ – Dynamics in smaller dimension.

$$\dot{\arcsin}\left(\frac{x_2}{r}\right) = \frac{\left(\frac{x_2}{r}\right)^{\cdot}}{\sqrt{1-\left(\frac{x_2}{r}\right)^2}} = \frac{\frac{p_2}{r} - \frac{x_2 p_r}{r^2}}{\sqrt{1-\left(\frac{x_2}{r}\right)^2}} =$$

Note that $\dot{\varphi} =$

$$= \frac{p_2 r^2 - x_2(p_1 x_1 + p_2 x_2)}{r^2\sqrt{r^2-x_2^2}} = \frac{p_2 x_1^2 - p_1 x_1 x_2}{r^2 x_1} = \frac{-L}{r^2}$$

# Lax matrix & integrability

Classical Lax matrix:

$L \in \mathrm{Mat}_{n\times n}(\mathbb{C}) \otimes \mathcal{F}(M) \iff L : M \longrightarrow \mathrm{Mat}_{n\times n}(\mathbb{C})$

{Quantization

$\mathrm{Mat}_{n\times n}(\mathbb{C}) \otimes \mathcal{A}$

**Lemma** Assume that $L : M \longrightarrow \mathrm{Mat}_{n\times n}(\mathbb{C})$ is such that

$\{L_1, L_2\} = [r_{12}, L_1] - [r_{21}, L_2]$ for some $r : M \longrightarrow \mathrm{Mat}_{n\times n}(\mathbb{C})^{\otimes 2}$.

Here $L_1 = L \otimes \mathbb{1}$, $L_2 = \mathbb{1} \otimes L$, $\{A_1, B_2\} := \sum_{i,j,k,\ell} \{A_{ij}, B_{k\ell}\} E_{ij} \otimes E_{k\ell}$.

Then $\forall s,t$: $\{\mathrm{Tr}(L^s), \mathrm{Tr}(L^t)\} = 0$.

**Proof:**

- $\{\mathrm{Tr}(A), \mathrm{Tr}(B)\} = \mathrm{Tr}(\{A_1, B_2\})$ ← (exercise)
- $\{A_1 \tilde{A}_1, B_2\} = A_1 \{\tilde{A}_1, B_2\} + \{A_1, B_2\} \tilde{A}_1$, sim. for the second component.
- $\{\mathrm{Tr}(L^s), \mathrm{Tr}(L^t)\} = \sum_{i=1}^{s} \sum_{j=1}^{t} \mathrm{Tr}\left(L_1^{i-1} L_2^{j-1} \{L_1, L_2\} L_1^{s-i} L_2^{t-j}\right) =$

$= s\cdot t\, \mathrm{Tr}\left(\{L_1, L_2\} L_1^{s-1} L_2^{t-1}\right) = s\cdot t\, \mathrm{Tr}\left(\left([r_{12}, L_1] - [r_{21}, L_2]\right) L_1^{s-1} L_2^{t-1}\right) =$

$= s\cdot t \Big( \underbrace{\mathrm{Tr}\left(r_{12} [L_1, L_1^{s-1} L_2^{t-1}]\right)}_{0} - \underbrace{\mathrm{Tr}\left(r_{21} [L_2, L_1^{s-1} L_2^{t-1}]\right)}_{0} \Big) = 0$

∎